CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

10.2 Kirchhoff's Laws

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

Question 1 Easy [1 mark]

The principles of conservation of which two quantities are associated with Kirchhoff's first and second Laws?

	First Law	Second Law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Kirchhoff's First Law (or current Law) states that current flowing into a junction must be equal to current flowing out of it
  - This is a consequence of charge conservation
  - It is well known in physics that charge will always be conserved. In the context of circuits this means that, since the current is the rate of flow of charge, the current flowing into a point must be the same as the current flowing out of that point
- Kirchhoff's Second Law (or voltage Law) states that the sum of all voltages around any closed loop in a circuit must equal the e.m.f. of the supply
  - This is a consequence of the principle of conservation of energy
  - This is due to the fact that work is done $W = VQ$ on charge carriers by the electric forces inside the circuit components meaning they either gain or lose electrical energy, depending upon the type of component

Question 2 Easy [1 mark]

In deriving a formula for the combined resistance of three different resistors in series, Kirchhoff's Laws are used.

Which physics principle is involved in this derivation ?

A the conservation of charge

B the direction of the flow of charge is from negative to positive

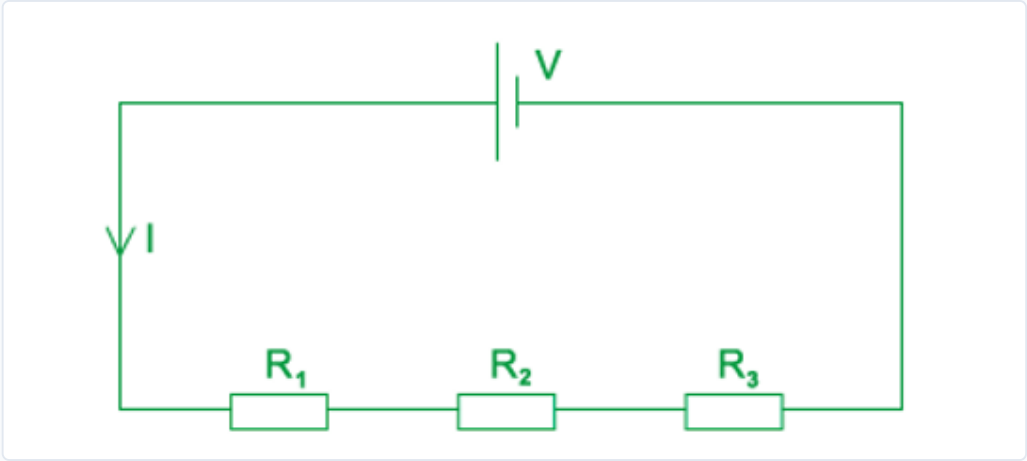
C the potential difference across each resistor is the same

D the current varies in each resistor, in proportion to the resistor value

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The physics principle involved in deriving Kirchhoff's Laws for 3 different resistors connected in series is the conservation of charge
- The shared e .m .f. source V causes a current I to flow through all three resistors, by the conservation of charge, this must be the same at all points in a series circuit
- From Ohm's Law $V = IR$
- The e. m .f. is split into V_1, V_2 and V_3 between the three resistors
  - $V_1 = IR_1 \quad V_2 = IR_2 \quad V_3 = IR_3$
- From Kirchoff's Second Law, we know that the voltages around a loop in a circuit add up to the e.m.f. of the supply
  - $V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = IR_1 + R_2 + R_3$
  - $\frac{V}{I} = R_1 + R_2 + R_3$
- From Ohm's Law, we know $\frac{V}{I}$ = total resistance of the circuit
  - $R_{total} = R_1 + R_2 + R_3$



B is incorrect because the flow of charge in a circuit is from the positive terminal of the battery to the negative terminal

C is incorrect because in series combination, the p .d. across different resistors is not the same since the resistance of each one is different

D is incorrect because in series combination, the current through each resistor is the same

Question 3 Easy [1 mark]

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Which statement is used in the derivation of this formula?

A the currents through the two resistors are equal

B the potential difference across each resistor is the same

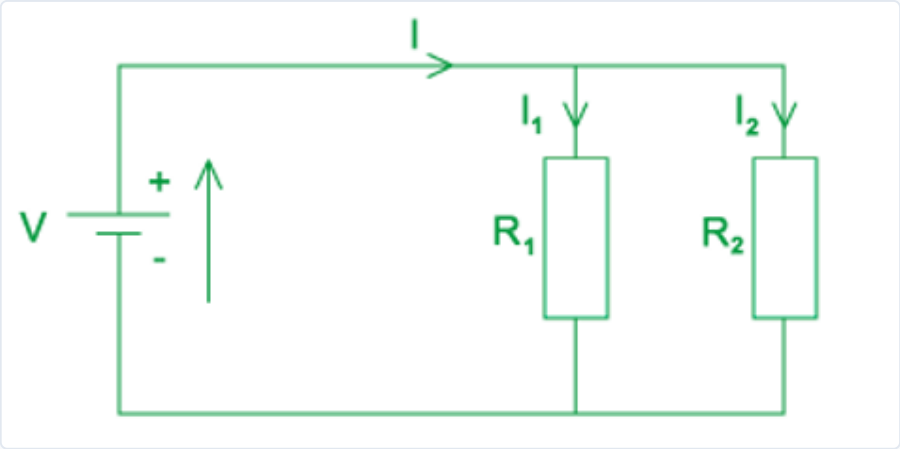
C the supply current is split between the two resistors in the same ratio as the ratio of their resistances

D the total power dissipated is the sum of the powers dissipated in the two resistors separately

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Each resistor in parallel has the same p .d. applied to it, but the total current is split amongst them
- Let the current through resistor R_1 be I_1 and the current through R_2 be I_2
- The p. d. across both R_1 and R_2 is equal to the supply voltage V
- Therefore from Ohm's Law
  - $I_1 = \frac{V}{R_1}$ and $I_2 = \frac{V}{R_2}$
- But from Kirchhoff's Current Law, we know the current entering a junction is equal to the current leaving the junction
  - Therefore, $I = I_1 + I_2$
- Substituting the values derived for I_1 and I_2 gives us'
  - $I = \frac{V}{R_1} + \frac{V}{R_2} = V \quad \frac{1}{R_1} + \frac{1}{R_2}$
- Rearranging gives us
  - $\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2}$
- From Ohm's Law, we know $\frac{V}{I} =$ total resistance of the circuit
  - $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2}$



A is incorrect because in parallel combination, the current through each resistor is different

C is incorrect because the supply current is split between the resistors according to the ratio between the supply p. d. and the resistance of each resistor $I = \frac{V}{R_1} + \frac{V}{R_2} = V \quad \frac{R_1 + R_2}{R_1 R_2}$

D is incorrect because the equation for power is not required in this derivation

Question 4 Easy [1 mark]

Each of Kirchhoff's two Laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** Law and names the quantity that is conserved ?

	Statement	Quantity
A	The algebraic sum of currents into a junction is zero	charge
B	The algebraic sum of currents into a junction is zero	energy
C	The e. m. f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	charge
D	The e .m .f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	energy

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Kirchhoff's First Law states: the algebraic sum of currents into a junction is zero
    - This is a consequence of charge conservation

B is incorrect because Kirchhoff's First Law is not a consequence of energy conservation

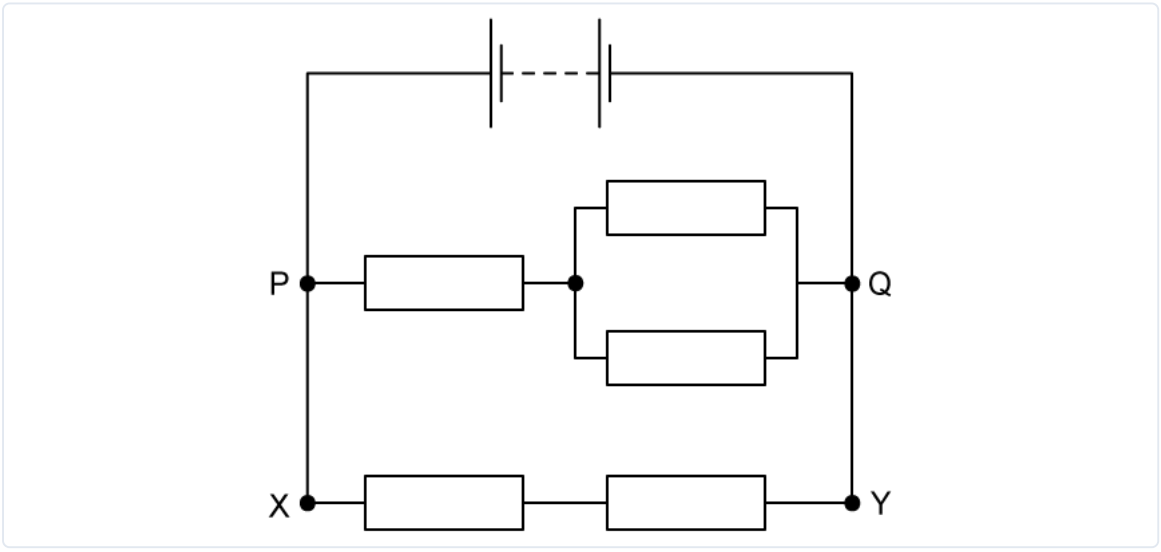
C & **D** are incorrect because these two options describe Kirchhoff's Second Law

Exam Tip: Remember Kirchhoff's First and Second Laws like this:

- Kirchhoff's First Law = Current Law = charge conservation $Q = I \quad t$
- Kirchhoff's Second Law = Voltage Law = energy conservation $W = VQ$

Question 5 Easy [1 mark]

A circuit consisting of five **identical** resistors is shown:



Which of the following statements is **false**?

A The potential difference between **P** and **Q** is equal to the potential difference between **X** and **Y**

B The total resistance between **X** and **Y** is equal to the sum of the resistances between **X** and **Y**

C The total resistance between **P** and **Q** is equal to the sum of the resistances along **P** and **Q**

D There is less total resistance between **P** and **Q** than there is between **X** and **Y**

Mark Scheme & Explanation:

*The false answer is **C** because:*

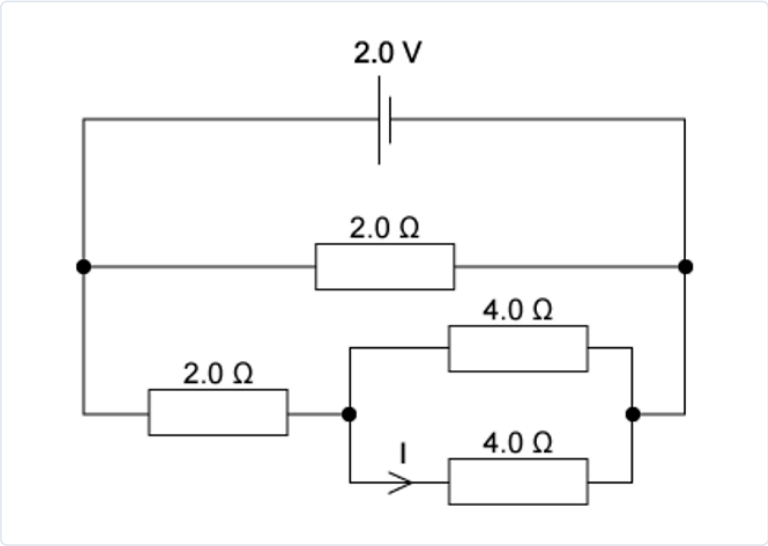
- The resistors between **P** and **Q** are not connected in series
    - Therefore, the total resistance is not equal to the sum of each resistance

A is true because the branch **PQ** is in parallel with the branch **XY**. Therefore, the potential difference across each branch is are equal **B** is true because the resistors between **X** and **Y** are connected in series; therefore, the total resistance $R_T = R_1 + R_2$

D is true because two of the resistors connected between **P** and **Q** are connected in **parallel**. Therefore, the total resistance here is **less** than one individual resistor. This means overall, there is less resistance along **PQ** than there is along **XY**

Question 6 Medium [1 mark]

A cell of e .m .f. 2.0 V and negligible internal resistance is connected to a network of resistors as shown.



What is the current *I* ?

A 0.25 A

B 0.33 A

C 0.50 A

D 1.5 A

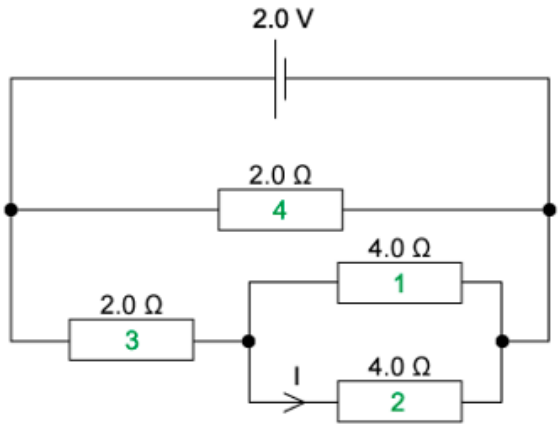
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- First, we must find the equivalent resistance of all the resistors combined
- We'll start with the two 4.0 Ω resistors, let's call these R_1 and R_2
- For resistors in parallel : $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
  - $\frac{1}{R_{1+2}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
  - So, $R_{1+2} = 2.0\ \Omega$
- Next, we'll add the combined R_1 and R_2 in series with R_3
- For resistors in series : $R = R_1 + R_2 + \cdots + R_n$

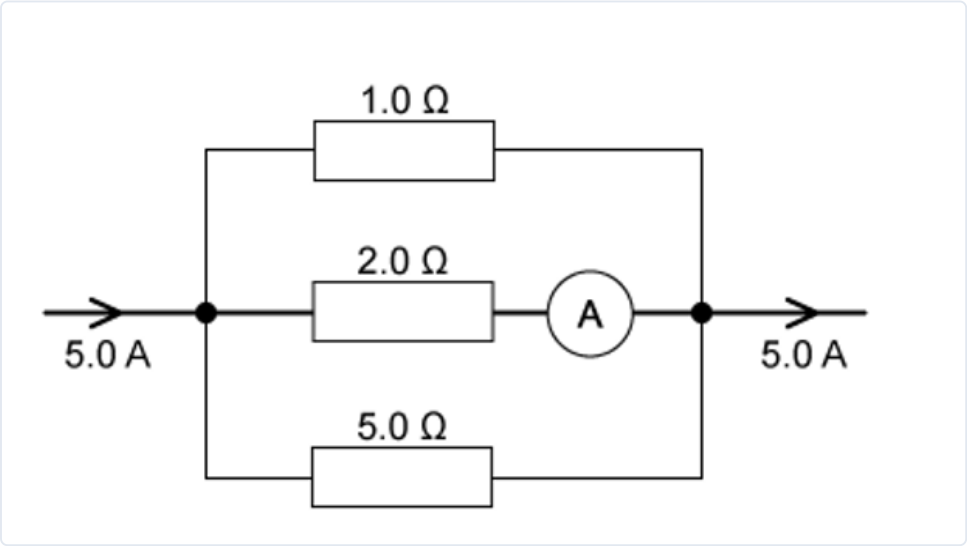
- So, $R_{1+2+3} = 2 + 2 = 4.0\ \Omega$

- And then, to get the total resistance, we'll add the combined 3 resistors in parallel with R_4
  - So, $\frac{1}{R_{total}} = \frac{1}{R_{1+2+3}} + \frac{1}{R_4} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$
- Therefore, the total resistance is $R_{total} = \frac{4}{3}\ \Omega$
- Now, we must find the value of the current *I* through R_2
- Ohm's Law states $I = \frac{V}{R}$
  - So, the current through the cell is equal to $I = \frac{V}{R} = \frac{2.0\ V}{\frac{4}{3}\Omega} = 1.5\ \ \text{A}$
- If we consider there to be three resistors of 2.0 Ω each, the current through each resistor will be $\frac{2}{6} \times 1.5 = 0.5\ \ \text{A}$
- Since the resistance is equal in both branches, the current will divide equally
  - So, $I = \frac{0.5}{2} = 0.25\ \ \text{A}$



Question 7 Medium [1 mark]

The diagram shows part of a current-carrying circuit. The ammeter has negligible resistance.



What is the reading on the ammeter ?

A 0.7 A

B 1.3 A

C 1.5 A

D 1.7 A

Mark Scheme & Explanation:

*The correct answer is **C** because:*

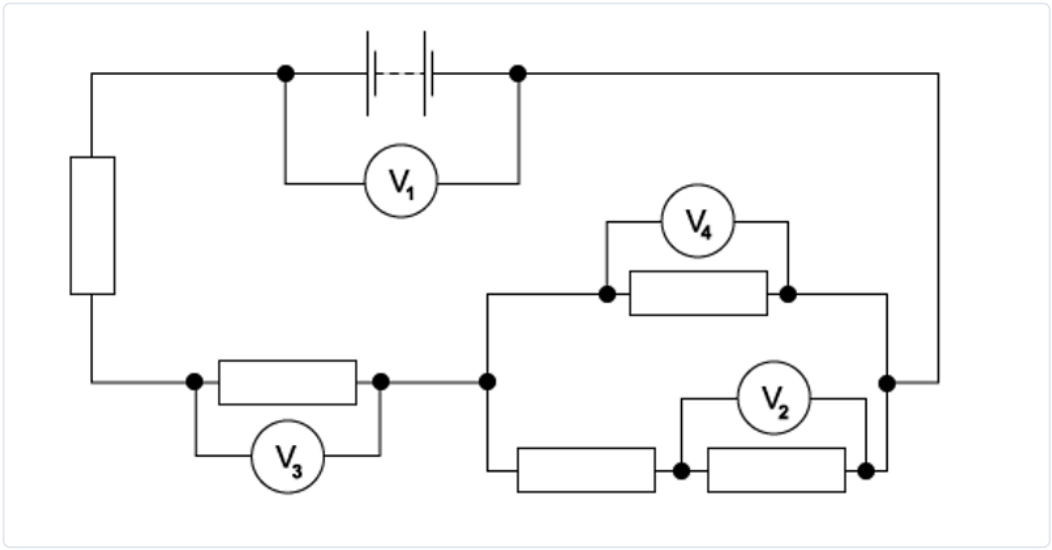
- Let the total resistance of the circuit be R
- For resistors in parallel : $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
  - $\frac{1}{R} = \frac{1}{1} + \frac{1}{2} + \frac{1}{5} = 1.7$
  - So, the total resistance is $R = \frac{1}{1.7} = 0.588 \quad \Omega$
- In other words, the three resistors in parallel is equivalent to a single resistor of resistance 0.588 Ω
- A current of 5.0 A flows through this resistor
  - So, the p. d. across the branches $= IR = 5.0 \times 0.588 = 2.94 \quad \text{V}$
- As the resistors are in parallel, the p. d. across each branch is the same

- $I = \frac{V}{R} = \frac{2.94}{2.0} = 1.47 \quad \text{A}$

- Therefore, the current through ammeter is 1.5 A

Question 8 Medium [1 mark]

In the circuit shown, all the resistors are identical.



The reading on voltmeter V_1 is 8.0 V and the reading on voltmeter V_2 is 1.0 V. What are the readings on the other voltmeters?

	Reading on voltmeter V_3 / V	Reading on voltmeter V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

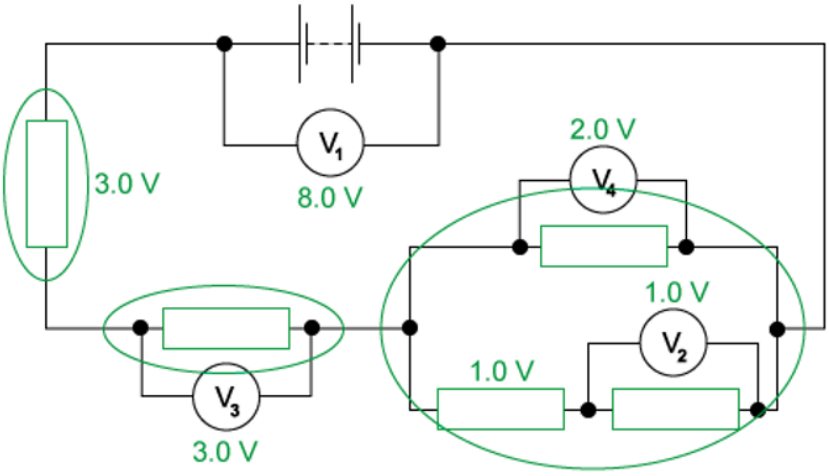
A
B
C
D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Let the resistance of each resistor be *R*
- The p .d. across a resistor is V_2 = 1.0 V

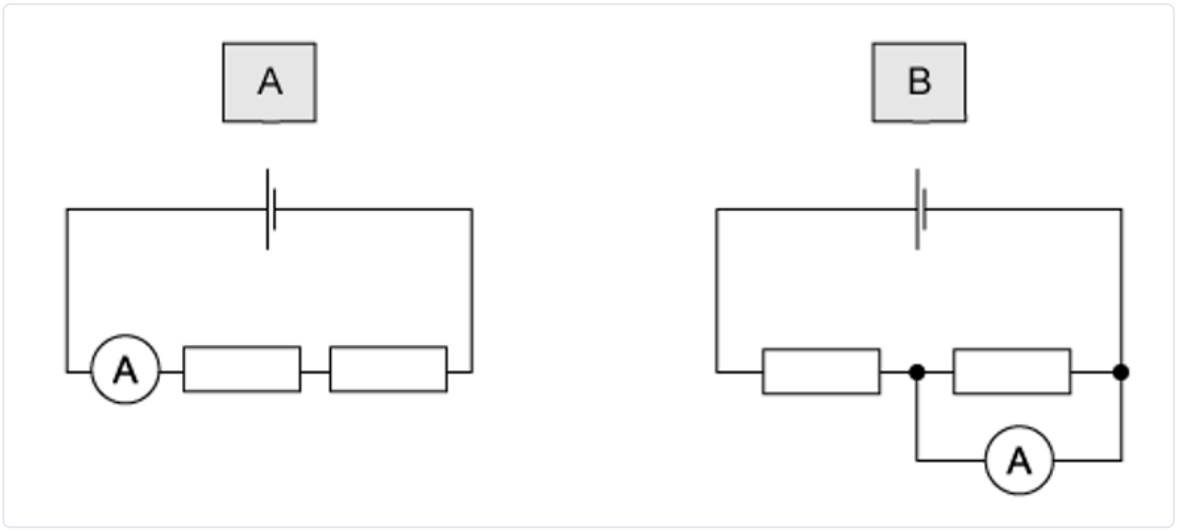
- The p. d. across the resistor just left of V_2 is also 1.0 V since the resistance is the same and they are in the same branch, so, the same current flows through them both
    - So, the total p. d. in the lower branch of the parallel combination is equal to = 1.0 + 1.0 = 2.0 V
- The total p .d. across any branch of a parallel combination is the same according to Kirchhoff's Second Law
    - Therefore, V_4 = 2.0 V
- The complete circuit may be described as a series combination of the two single resistor and the parallel combination (as shown in the diagram)
- The p .d. across the vertical resistor is the same as V_3 since they have the same resistance.
    - V_1 = total e. m .f in circuit = 8.0 V
- The sum of p .d. in any loop is equal to the e. m .f in the circuit
    - $V_3 + V_3 + 2.0 = 8.0$ V
    - $2V_3 = 6.0$ V
- So, V_3 = 3.0 V

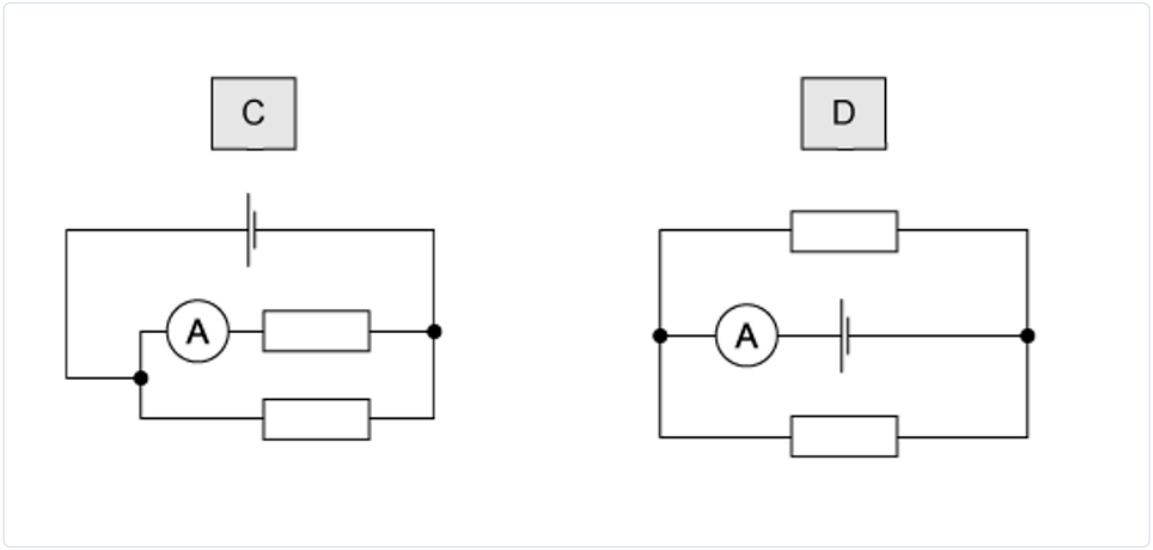


Question 9 Medium [1 mark]

A cell, two resistors of equal resistance and an ammeter are used to construct four circuits. The resistors are the only parts of the circuits that have resistance.

In which circuit will the ammeter show the greatest reading ?





A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Ohm's Law states $V = IR$, so, the current in the whole circuit is $I = \frac{V}{R_T}$

    - Where R_T is the combined resistance of the resistors

- Let the resistance of a resistor = R and the e .m. f of the supply = E
- In circuit **D**, the 2 resistors are in a parallel combination, so the combined resistance is
    - $\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$, therefore, $R_T = \frac{R}{2}$
- Therefore, the current in the circuit is $I = \frac{E}{\frac{R}{2}} = \frac{2E}{R}$
- This is the same current as circuit **C**, however, at a junction, the current splits
- In circuit **D**, the ammeter is connected directly to the positive terminal, so it reads the value of the current flowing in the whole circuit, whereas the current in **C** will read half the total current
    - Therefore, the ammeter in circuit **D** will have the greatest reading

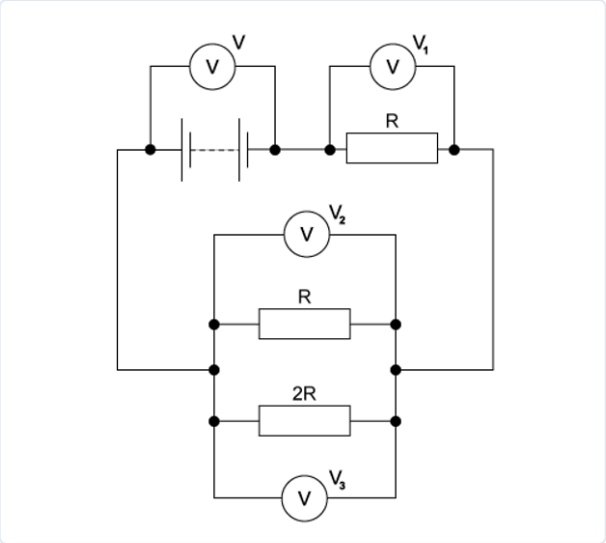
A & **B** are incorrect because these two circuits are in a series arrangement, the current will be the lower than in a parallel arrangement:

- Circuit **A**: $R_T = R + R = 2R$, so, the current $I = \frac{E}{2R}$
- Circuit **B**: $R_T = R + 0 = R$, so, the current $I = \frac{E}{R}$

C is incorrect because in circuit **C** the current splits at a junction, so, the ammeter in circuit **C** gives only half the total current in the circuit

Question 10 Medium [1 mark]

The diagram shows a circuit with four voltmeter readings V, V_1, V_2 and V_3.



Which equation relating the voltmeter readings must be true?

A $V = V_1 + V_2 + V_3$

B $V + V_1 = V_2 + V_3$

C $V_3 = 2(V_2)$

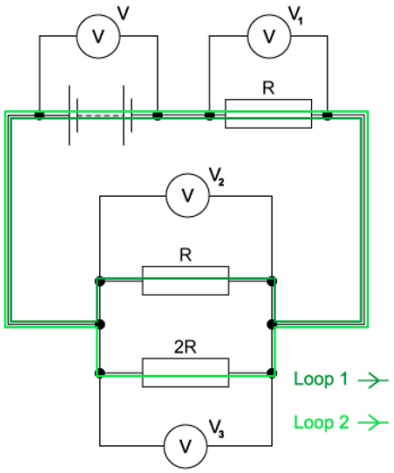
D $V - V_1 = V_3$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

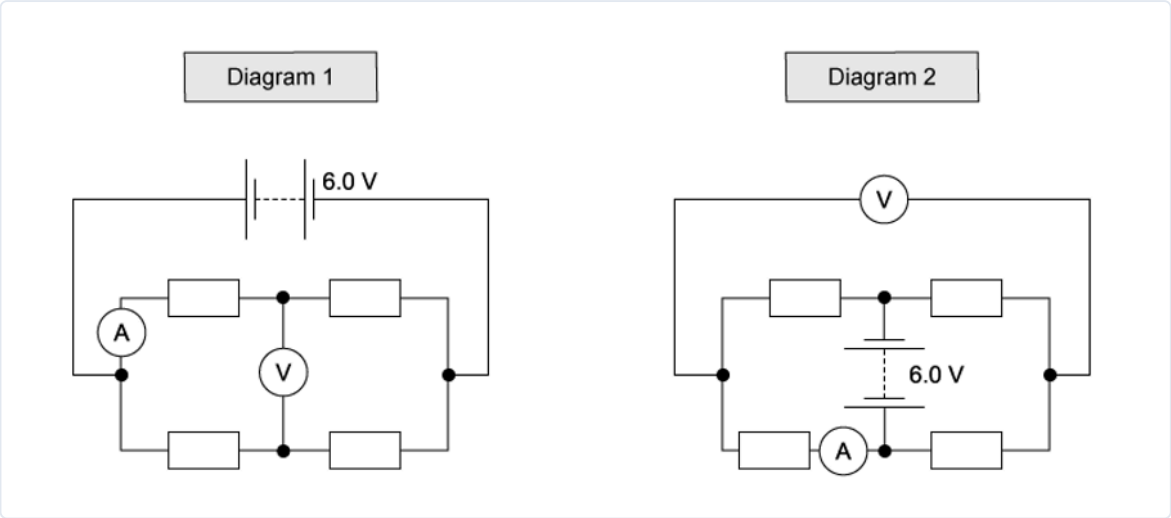
- The voltmeters are used to give the potential difference across the resistors. The voltmeter reading V gives the e.m.f of the battery
- The resistors connection in the circuit forms two different loops as shown in the diagram below (the voltmeters are not considered)
  - Loop 1: battery — resistor R (in parallel) — resistor R
  - Loop 2: battery — resistor 2R (in parallel) — resistor R
- From Kirchhoff's Laws, the sum of p. d. in any loop should be equal to the e. m .f in the circuit
  - $V = V_2 + V_1$

- $V = V_3 + V_1$
- $V - V_1 = V_3$



Question 11 Medium [1 mark]

When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A, and the voltmeter reads zero.



The resistors and meters are reconnected to the supply, as shown in diagram 2.

What are the meter readings in diagram 2?

	Voltmeter reading / V	Ammeter reading / A
A	0	1.0
B	3.0	0.5
C	3.0	1.0
D	6.0	0

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- In diagram 1, the voltmeter reads zero because the potential difference across it is zero, meaning that both of its junctions are at the same potential
- From Kirchhoff's First Law, current splits at a junction

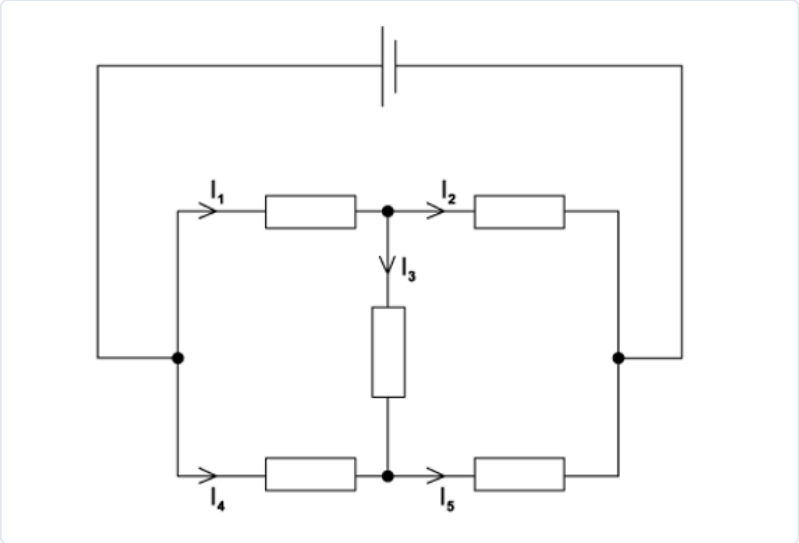
- Since the resistors are identical, the current splits equally into the two branches
    - The ammeter reads a current of 1.0 A, so the total current in the circuit is 1.0 + 1.0 = 2.0 A
- Now, let's consider diagram 2: current flows from the positive terminal of the supply and is split equally at a junction, so, the ammeter still reads the current that has been split equally into two
    - This current is equal to $\frac{2.0}{2} = 1.0 \quad \text{A}$
- From the junction connected to the positive terminal of the supply, the current splits at the bottom junction and rejoins again at the top junction to return to the negative terminal
- Since the same amount of current flows through identical resistors the potential of the left and right junctions would be the same
    - Therefore, the potential difference across the voltmeter is zero

B & **D** are incorrect because the total current in diagram 1 and 2 is 2.0 A so when the current splits at the junction, the reading on both ammeters is $\frac{2.0}{2} = 1.0 \quad \text{A}$

C is incorrect because to detect a potential difference, the voltmeter would need to be connected across one component at a time

Question 12 Medium [1 mark]

The diagram shows currents I_1, I_2, I_3, I_4 and I_5 in different branches of a circuit.



Which one of the following is correct?

A $I_1 = I_2 + I_3$

B $I_2 = I_1 + I_3$

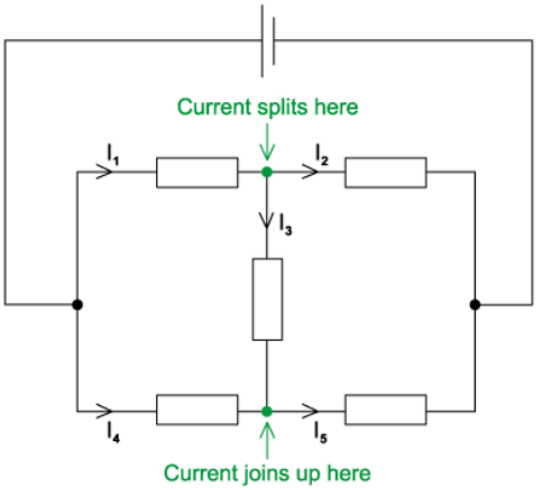
C $I_3 = I_4 + I_5$

D $I_4 = I_5 + I_3$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Kirchhoff's First Law states that the total current entering a junction is equal to the total current leaving the same junction
- In other words the algebraic sum of all the current entering and leaving a junction must be equal to zero, so : $\sum I_{IN} = \sum I_{OUT}$
- At the top junction, indicated in the diagram, I_1 splits into I_2 and I_3
  - $I_1 = I_2 + I_3$
- At the bottom junction, indicated in the diagram, I_3 and I_4 join up to become I_5
  - $I_5 = I_3 + I_4$



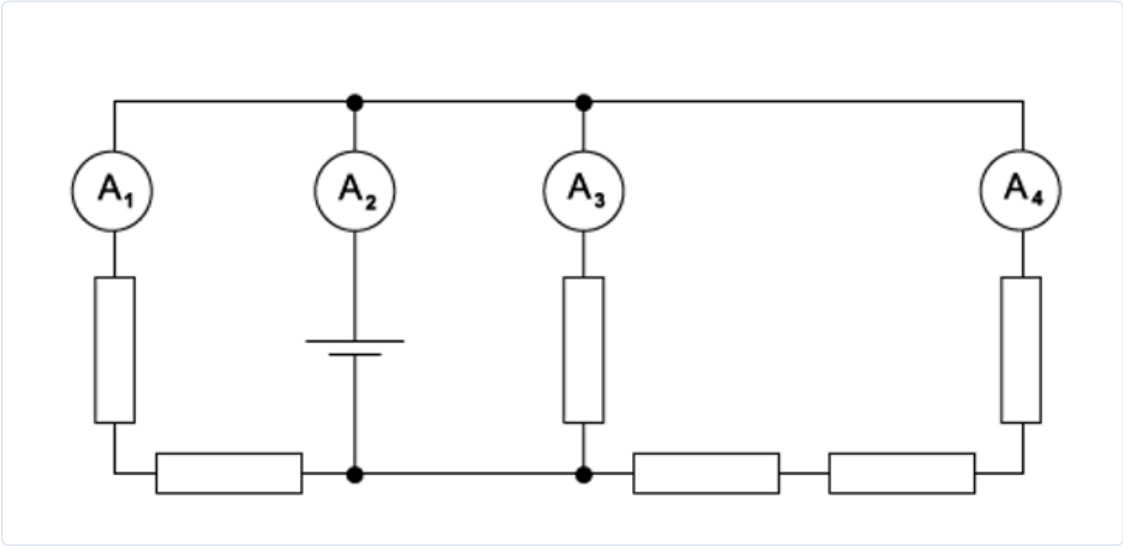
B is incorrect because current I_2 is equal to $I_2 = I_1 - I_3$

C is incorrect because current I_3 is equal to $I_3 = I_5 - I_4$

D is incorrect because current I_4 is equal to $I_4 = I_5 - I_3$

Question 13 Medium [1 mark]

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



The reading on ammeter A_1 is 0.6 A.

What are the readings on the other ammeters?

	Reading on ammeter A_2 / A	Reading on ammeter A_3 / A	Reading on ammeter A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

A

B

C

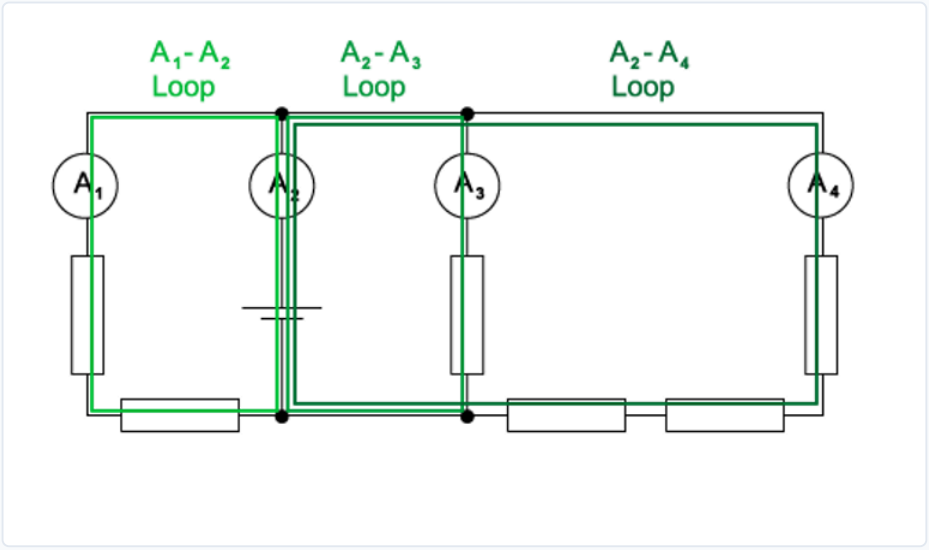
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

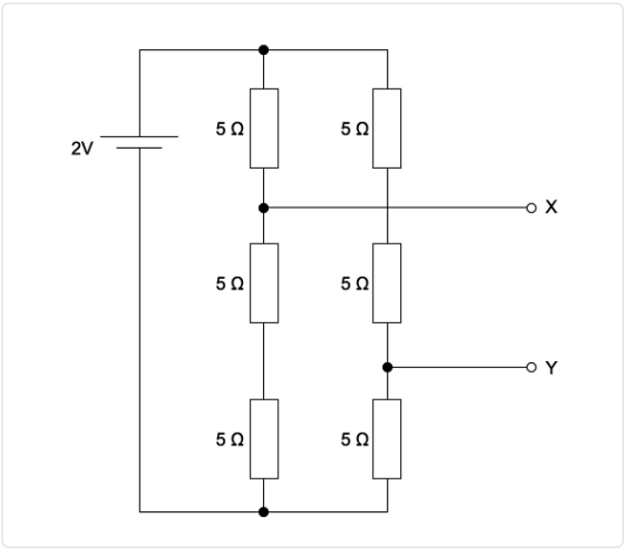
- Current flows from the positive terminal of the battery, so, the current in A_2 should have the highest value
- Let the resistance of each resistor be *R*

- The left-hand side of the circuit, in the (A_1 - A_2) loop, there are two resistors in series, so their total resistance is $R + R = 2R$
    - The current in the (A_1 - A_2) loop = 0.6A, as read by ammeter A_1
- So, from Ohm's Law $V = IR$, the potential difference in the (A_1 - A_2) loop is equal to $V = 0.62R = 1.2 \quad R$
- From Kirchhoff's Second Law, the potential difference is the same as the e .m .f in the circuit for any loop, so, the e. m .f. in the circuit is 1.2*R*, and this is also the potential difference across all loops
- Consider the (A_2 - A_3) loop:
    - Current in A_3 $= \frac{V}{R} = \frac{1.2 \quad R}{R} = 1.2 \quad \text{A}$
- Similarly, consider the (A_2 - A_4) loop:
    - Current in A_4 $= \frac{1.2 \quad R}{3R} = 0.4 \quad \text{A}$
- Therefore, current in A_2 = total current = 0.6 + 1.2 + 0.4 = 2.2A



Question 14 Hard [1 mark]

Six resistors, each of resistance 5 Ω, are connected to a 2 V cell of negligible internal resistance.



What is the potential difference between terminals X and Y ?

A $\frac{2}{3}$ V

B $\frac{8}{9}$ V

C $\frac{4}{3}$ V

D 2 V

Mark Scheme & Explanation:

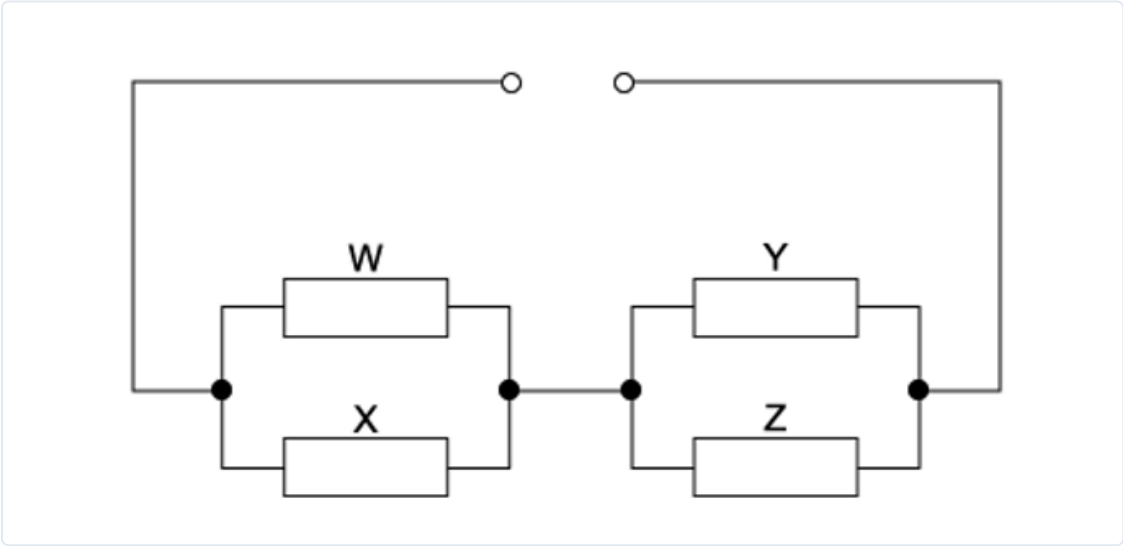
*The correct answer is **A** because:*

- To obtain the potential difference between terminals X and Y, we first need to know the values of the potential at X and Y
  - The positive terminal of the cell can be taken to be at a potential of 2 V and the negative terminal to be at a potential of 0 V
- It can be seen that the circuit consists of 2 loops, with a junction at the top and one at the bottom
  - Since there are no components connected between the upper junction and the positive terminal of the cell, the upper junction must be at a potential of 2 V, and the lower junction must be at a potential of 0 V
- Therefore, in any loop, going from up to down, the potential decreases across each component until it is zero at the lower junction

- From Kirchhoff's Law, the sum of p .d. in any loop is equal to the e. m .f in the circuit
- Consider the first loop (where terminal X is connected):
- From the potential divider equation:
  - p. d. across the upper resistor $= \frac{R_1}{R_1 + R_2 + R_3} \times E = \frac{5}{5+5+5} \times 2 = \frac{2}{3} \quad V$
  - Note that this is the potential difference across the upper resistor. It is also the p. d. across each resistor since they all have the same resistance
- Since the upper junction is at a potential of 2 V, the p .d. at terminal X is equal to $= 2 - \frac{2}{3} = \frac{4}{3} \quad V$
- Consider the second loop (where terminal Y is connected):
- From the upper junction to terminal Y, there are 2 resistors and the p. d. across each is $\frac{2}{3} \quad V$
  - So, the p. d. at terminal Y $= 2 - 2\frac{2}{3} = \frac{2}{3} \quad V$
- Therefore, the p .d. between X and Y $= \frac{4}{3} - \frac{2}{3} = \frac{2}{3} \quad V$

Question 15 Hard [1 mark]

Four resistors of equal value are connected as shown.



How will the powers to the resistors change when resistor W is removed?

A the powers to X, Y and Z will all increase

B the power to X will decrease and the powers to Y and Z will increase

C the power to X will increase and the powers to Y and Z will decrease

D the power to X will increase and the powers to Y and Z will remain unaltered

Mark Scheme & Explanation:

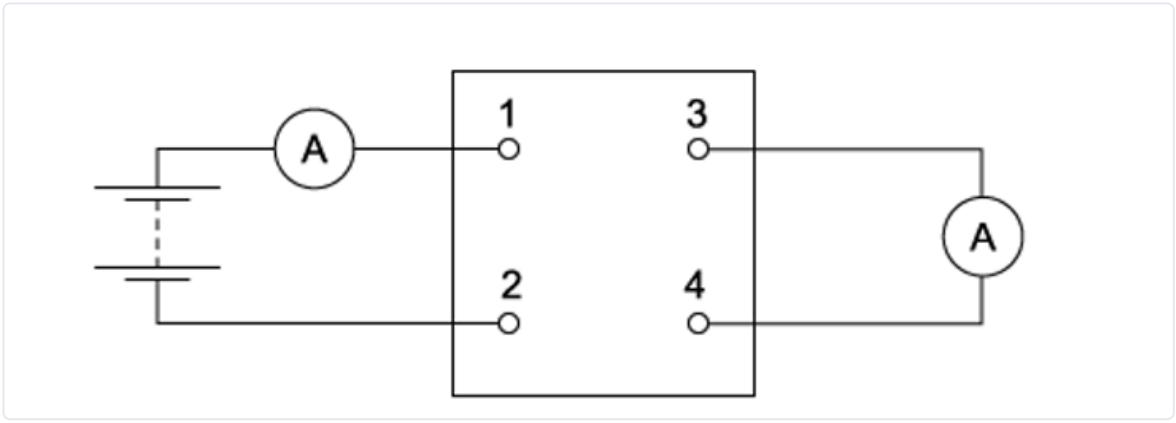
*The correct answer is **C** because:*

- Power dissipated in a resistor $= I^{2} R$
- Let's say the resistance of each resistor is *R*
- Two resistors in series causes the overall resistance to increase: *R* + *R* = 2*R*
- Two resistors in parallel causes the overall resistance to decrease : $\frac{1}{R} + \frac{1}{R}^{-1} = \frac{R}{2}$
- Since the 4 resistors have the same resistance in the connection shown, the p .d. across each parallel connection will be the same
    - Overall the resistance in the circuit is $= \frac{R}{2} + \frac{R}{2} = R$
- Let's say the e .m .f. in the circuit is *E*, so, the current in the circuit is $I = \frac{E}{R}$

- The total current splits at the parallel combination
  - Power dissipated in X $= \frac{1}{2}^2 R = \frac{I^2 R}{4} = 0.25 \quad I^2 R$
  - This is the power dissipated in all the resistors
- When W is removed, the overall resistance of the parallel combination of W and X changes from 0.5*R* to *R* as only X is left
- Total resistance in new circuit = *R* + 0.5*R* = 1.5*R*
  - Current in circuit $= \frac{E}{1.5 \ R} = \frac{2}{3}I$
- The overall current in the circuit will also change — it decreases since the overall resistance in the complete circuit has increased. This current flows through X but is split into half at the combination of Y and Z, so, the current through Y and Z is half that through X, so:
  - Power dissipated in X $= \frac{2}{3}I^2 \quad R = \frac{4}{9}I^2 \quad R = 0.44 \quad I^2 R$
- Half the current flows through Y and Z, and the same power is dissipated through both resistors, so:
  - Power dissipated in Y $= \frac{1}{2}\frac{2}{3}I^2 \quad R = \frac{1}{9}I^2 R = 0.11 \quad I^2 R$
- Therefore, the power in X increases and the power in Y and Z decreases

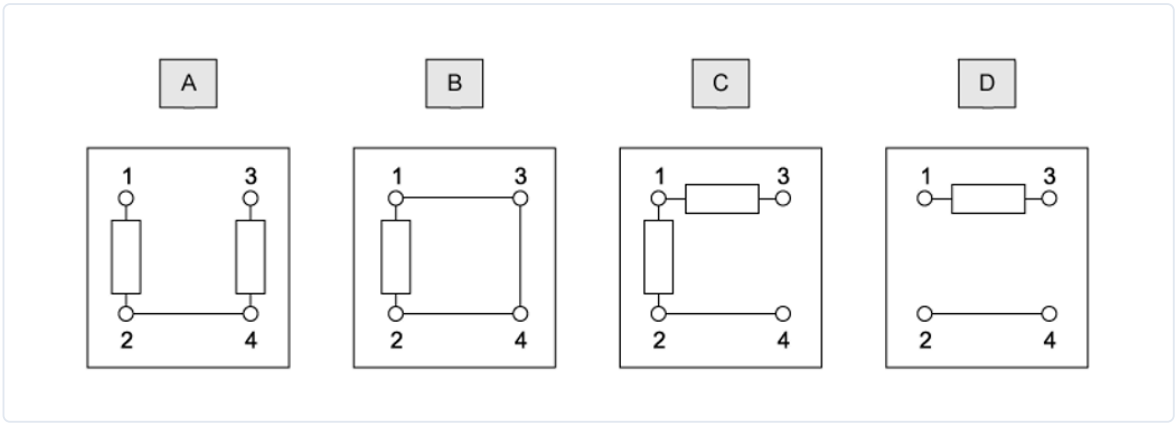
Question 16 Hard [1 mark]

The diagram shows a four-terminal box connected to a battery and two ammeters.



The currents in the two meters are identical.

Which circuit, within the box, will give this result ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The ammeter connected close to the battery gives the total current in the circuit while the ammeter connected between terminals 3 and 4 gives the current passing through terminal 3 to 4
- The current in this one will only be equal to the one close to the battery only if all components are connected in series in the whole circuit, including within the box
- Consider circuit **D** :

- The 2 ammeters are in series with the resistor between terminals 1 and 3
- So, the same current flows through components connected in series
- Therefore, option **D** is correct

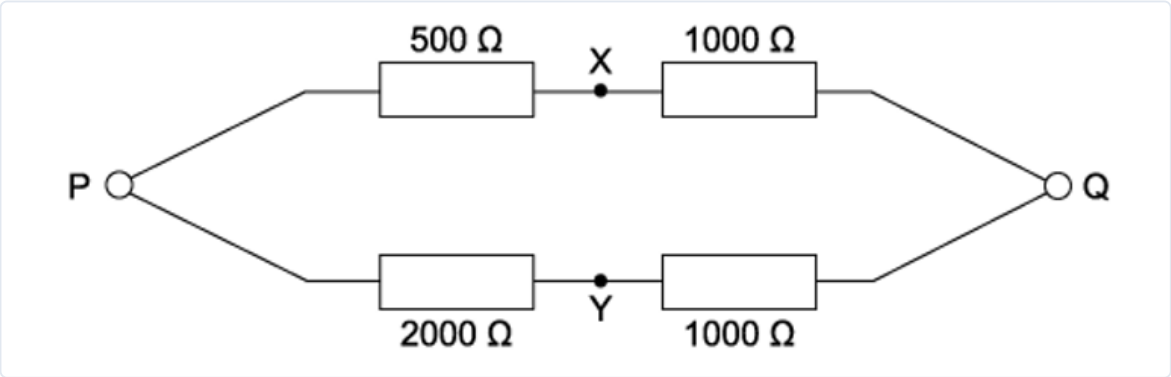
A is incorrect because the section from terminal 2 to 4 and finally to 3 is short-circuited (since 1 and 3 are not connected), so, no current flows through the ammeter

B is incorrect because the ammeter connected to 3 and 4 is short-circuited since the wires (which have negligible resistance) are connected between 1 to 3, 3 to 4 and 4 to 2

C is incorrect because a parallel connection is formed between the 2 resistors between 1 to 3 and 1 to 2, so, the current splits up at junction 1. The second ammeter is connected in series with one of the resistors, so it only gives the current flowing in one resistor

Question 17 Hard [1 mark]

A p. d. of 12 V is connected between P and Q.



What is the p. d. between X and Y?

A 0 V

B 4 V

C 6 V

D 8 V

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The statement "p. *d. of 12 V is connected between P and Q* " means that the difference in potential between P and Q is 12 V
- So, any potential can be assigned to P and Q as long as the difference is 12 V
    - Let's assign a potential of 0 V at P and 12 V at Q
- So, when we go from P to Q, the potential should increase and point X would be at a potential between 0 and 12 V
- The same is true for point Y; hence, it can be concluded that points P, Q, X and Y are all at different potentials
- The circuit consists of a parallel combination of resistors
    - So, the total p. d. across the upper and lower parts of the circuit are both 12 V
- Now, both the upper and lower part of the circuit each consist of two resistors in series
- For series connection of two resistors R_1 and R_2, the p. d. across R_1 is given by the potential divider equation

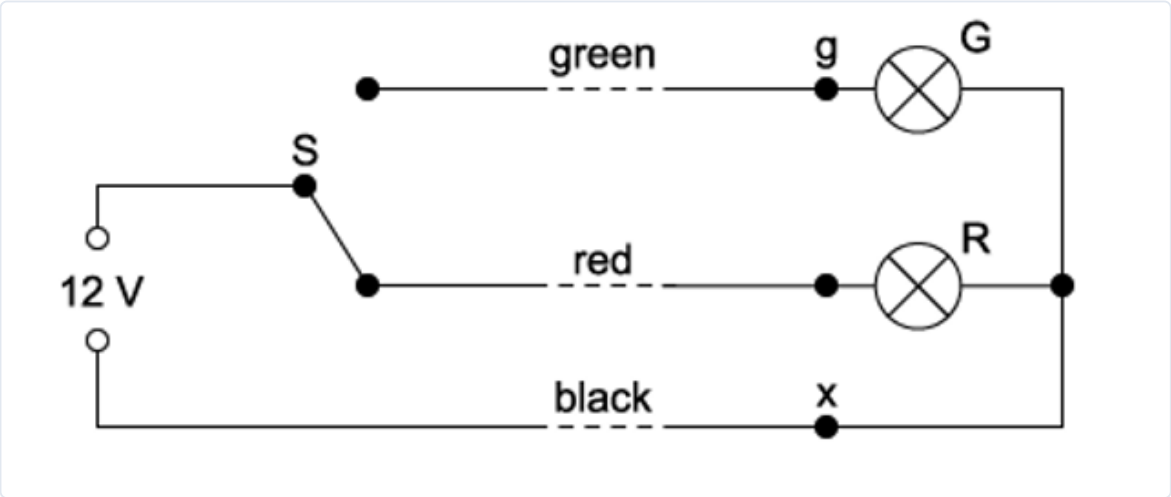
- $V = \frac{R_1}{R_1 + R_2} \times E$ where *E* is the e. m. f. of the supply

- The potential at point X = p. d. across 500 Ω in the upper part
  - Potential at point X $= \frac{500}{500 + 1000} \times 12 = 4 \quad \text{V}$
- Similarly, the potential at point Y = p. d. across 2000 Ω in the lower part
  - Potential at point Y $= \frac{2000}{2000 + 1000} \times 12 = 8 \quad \text{V}$
- So, the p .d. between X and Y is the difference in potential at point X and point Y
  - Therefore, p .d. between X and Y = 8 – 4 = 4 V

Exam Tip: Note that 0 V could be assigned at Q and 12 V at P, but the process discussed would have to be reversed

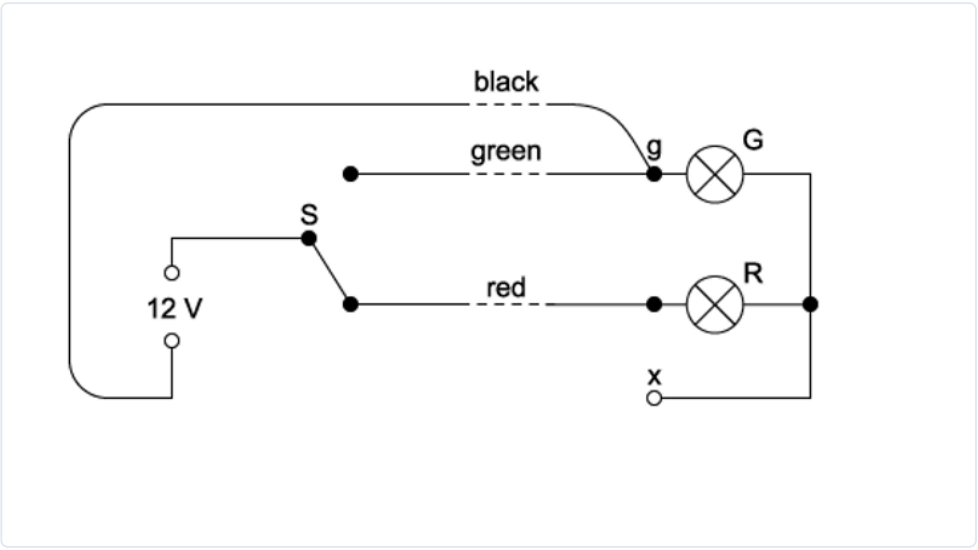
Question 18 Hard [1 mark]

The diagram shows the circuit for a signal to display a green or a red light. It is controlled by the switch S.



The signal is some way from S to which it is connected by a cable with green, red and black wires. At the signal, the green and red wires are connected to the corresponding lamp and the black wire is connected to a terminal x to provide a common return. The arrangement is shown correctly connected and with the switch set to illuminate the red lamp.

During maintenance, the wires at the signal are disconnected and, when reconnected, the black wire is connected in error to the green lamp (terminal g) instead of terminal x. The red wire is connected correctly to its lamp and connections at S remain as in the diagram.



When the system is tested with the switch connection to the red wire, what does the signal show ?

A the green lamp illuminated normally

B the red lamp illuminated normally

C the red and green lamps both illuminated normally

D the red and green lamps both illuminated dimly

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The lamps are both lit dimly as they are connected in series
- From Kirchhoff's Second Law, the total e.m.f. is now shared across two lamps in series
- Each of the lights now receive a voltage lower than 12 V
- Previously, all the 12 V was being provided to a single lamp which would cause it to be brighter

A & **B** are incorrect because both lamps are connected in series; hence they will both be lit

C is incorrect because both lamps are connected in series, but now the voltage is shared between them; hence they cannot have the same brightness as the single lit lamp previously